RELATIVE FINITE PRESENTATION COINCIDES WITH ABSOLUTE OVER A FINITELY PRESENTABLE BASE MAP, AND DETECTS THE BASE MAP ITSELF

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 16452-16463 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 16465-16470, the complete author proof (4 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. BORDERLINE for Chief: proof is a see-the-discussion pointer whose substance lives in the referenced definition block; accepted because the two clauses are exactly the definitional dictionary (no hidden argument), flagged in case the lane should treat discussion-pointers as rejected. Dictionary card used by this wave's module-closure siblings L16182/L16183.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. More on Algebra, definition relatively-finitely-presented and its accompanying dictionary discussion; the chapter's local-finite-type hypothesis for the presentation comparison.

Source reference. Original lemma 05GZ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 16470–16470. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes which is locally of finite type. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module.*

(1) *If f is locally of finite presentation, then $\mathcal{F}$ is of finite presentation relative to S if and only if $\mathcal{F}$ is of finite presentation.*
(2) *The morphism f is locally of finite presentation if and only if $\mathcal{O}_X$ is of finite presentation relative to S.*

Proof. Follows immediately from the definitions, see discussion following More on Algebra, Definition 05GZ. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.